

PEO vs Dashboard Stat Cards — Correlation Explainer

Audience: Non-IT faculty (Dean, Chairpersons), accreditation reviewers

Purpose: Explain why PEO percentages and Dashboard percentages differ even though they come from the same alumni records

Companion to: [PEO-EXPLAINER.md](#) (caveman edition of PEO logic) and [PEO-IMPLEMENTATION-PLAN.md](#) (technical spec)

TL;DR

Card type	Asks	Numerator	Denominator
Dashboard cards	"What % of alumni meet <i>this one</i> condition?"	One condition	Varies (all / employed / CE+EE)
PEO cards	"What % attain <i>this objective</i> (multiple paths)?"	OR of several criteria	Total respondents (default)

PEO cards are **broader numerators on a stable denominator**. Dashboard cards are **narrow numerators on whichever denominator makes the KPI meaningful**.

Side-by-Side: What Each Card Actually Computes

Dashboard stat cards (bottom row of the dashboard)

Card	Numerator	Denominator	Pass condition
Board Passers	<code>has_license = 1</code>	Alumni in CE + EE only (CpE excluded — no PRC board)	single condition
Employed	<code>is_employed = 1</code>	All alumni in scope	single condition
Field-Related Work	<code>is_employed = 1 AND job_relevance ∈ {highly, moderately}</code>	Employed alumni only	single condition
Supervisory / Managerial	<code>is_employed = 1 AND job_level ∈ {supervisory, managerial}</code>	Employed alumni only	single condition

These are **single-axis KPIs** — each measures one thing.

PEO stat cards (top row of the dashboard)

Card	Numerator	Denominator	Pass condition
PEO 1 — Professional Competence	passing PEO 1	All in scope (denominatorMode='total')	is_employed=1 AND (job-related OR has-license/cert OR supervisory) — OR of 3 criteria
PEO 2 — Ethics & Social Responsibility	passing PEO 2	All in scope	is_employed=1 AND (job-related OR community OR awards) — OR of 5 sub-criteria
PEO 3 — Innovation & Sustainability	passing PEO 3	All in scope	is_employed=1 AND (job-related OR research OR innovation-sector) — OR of 6 sub-criteria
Outcomes (Field-Aligned Employment)	aligned alumni	All in scope	is_employed=1 AND job-related — same numerator as Field-Related card, different denominator

PEO cards are **OR-composite scores** — many ways to pass.

Worked Comparisons (Numbers from the Sample Screenshot)

Sample scope: **688 total respondents** · **570 employed** · **476 in CE+EE**.

1. PEO 1 (76.2% on 688) vs Field-Related (78.4% on 570)

- Numerators differ. PEO 1 counts anyone *employed who is job-related* **OR** licensed **OR** supervisory — a **broadier** numerator than Field-Related.
- But PEO 1 divides by **688** (everyone), while Field-Related divides by **570** (employed only).
- Field-Related's smaller denominator pushes its rate up; PEO 1's broader numerator partly compensates → they end up close, but PEO 1 is slightly lower.

2. Outcomes 65.0% (447/688) vs Field-Related 78.4% (447/570)

- **Same 447 numerator.** The difference is 100% denominator choice.
- Outcomes uses *total respondents* (the accreditation-defensible default).
- Field-Related uses *employed only* (the operational HR-style KPI).

3. PEO 1 76.2% vs Board Passers 63.0%

- **Not directly comparable.**
- Board Passers excludes CpE entirely (476 denominator).
- PEO 1 includes CpE in its 688.
- Having a license is just *one of three* paths to PEO 1.

4. Why are PEO numbers all lower than Employed (82.8%)?

Every PEO has the **employment gate** built in. A PEO % can never exceed the Employed %. Outcomes (65.0%) $\leq$ Employed (82.8%) for the same reason.

Quick Rule of Thumb

Dashboard cards = single survey question, % of relevant population.

PEO cards = *attainment of an objective* satisfied by multiple alternative criteria, % of total respondents (so accreditors can compare across cohorts on a stable base).

If you toggle the PEO accordion's **Denominator** → "**Employed only**", the PEO percentages will jump because the same passing count is divided by 570 instead of 688.

What "As-of Year" Does in the PEO Filter

asOfYear is **the reference year used to assign each alumnus to a cohort** for the Outcomes card. It does **not** filter rows — it relabels them.

Default: server-side `new Date().getFullYear()` (currently **2026**).

The cohort boundaries are computed as $Y = \text{asOfYear}$:

Cohort	Window	Years since graduation
Recent Graduate	<code>year_graduated >= Y - 2</code>	0–2
Mid-Career	<code>year_graduated BETWEEN Y - 5 AND Y - 3</code>	3–5
Established	<code>year_graduated BETWEEN 2018 AND Y - 6</code>	6+

Example with default $Y = 2026$

- **Recent** = grads from **2024–2026**
- **Mid** = grads from **2021–2023**
- **Established** = grads from **2018–2020**

If you set **asOfYear** = **2025**, the same alumnus who graduated in 2023 moves from "Mid" to "Recent" — because in 2025 they were only 2 years out, not 3.

Why expose it as an override?

1. **Reproducing a past report.** Reviewers may ask: "*Show me the cohort breakdown as it would have looked when you filed in 2024.*" Setting **asOfYear=2024** rebuilds the cohort splits as they stood that year.
2. **Year-over-year reproducibility.** Today's "Established" cohort drifts every January 1. Pinning **asOfYear** freezes the math.
3. **Accreditation evidence trail.** The PDF/DOCX export prints the **asOfYear** on the cover page so the document is self-describing.

What **asOfYear** does NOT affect

- PEO 1, PEO 2, PEO 3 attainment percentages — they don't use cohorts.
- Year From / Year To filters — those exclude rows; **asOfYear** only relabels them.
- The schema floor of 2018 — Established is always clamped at **>= 2018**.

So: leave it blank for "current snapshot," set it explicitly when reproducing a historical report or freezing numbers for a formal submission.

Caveman Edition

Same explanation, fewer big words.

Why two rows of percent cards confuse people

Top row of cards = **PEO numbers** (the accreditation promises). Bottom row of cards = **Dashboard numbers** (regular survey stats).

They count from the same alumni list. But they ask different questions, so numbers different. That is OK. That is by design.

Caveman: Dashboard cards

Card	Caveman question
Board Passers	"How many CE and EE alumni have license?" (CpE no take board exam, so we skip them.)
Employed	"How many alumni have job?"
Field-Related Work	"Of those who have job, how many job match their degree?"
Supervisory	"Of those who have job, how many are boss?"

One question. One simple number. Bottom of denominator changes — sometimes "all alumni," sometimes "only employed alumni."

Caveman: PEO cards

Card	Caveman question
PEO 1 — Professional Competence	"How many alumni became GOOD at job?" (Pass if employed AND job-related OR has license OR is boss.)
PEO 2 — Ethics & Social Responsibility	"How many alumni became GOOD person who help community?" (Pass if employed AND job-related OR community work OR awards.)
PEO 3 — Innovation & Sustainability	"How many alumni became MAKER of new things?" (Pass if employed AND job-related OR research OR work in tech/innovation sector.)

Card	Caveman question
Outcomes	"How many alumni doing job that matches their school degree?" (Same as Field-Related card but uses ALL alumni as bottom number.)

Many ways to pass. Bottom of denominator always = ALL alumni in scope (default). This makes accreditation people happy because the bottom number doesn't move around.

Caveman: Why numbers different

Same numerator, different bottom

Outcomes 65.0% (447/688) and **Field-Related 78.4% (447/570)** = SAME 447 people on top. Just different bottom number.

- $447 \div 688$ (everyone) = 65%
- $447 \div 570$ (only employed) = 78%

Same people. Different math. Different percentage.

Different paths to pass

PEO 1 counts anyone who is employed AND (job-related OR licensed OR boss). **Field-Related** counts anyone who is employed AND job-related.

PEO 1 has more ways to pass → bigger top number. But PEO 1 divides by everyone (688) → smaller percentage push. End result: PEO 1 (76.2%) and Field-Related (78.4%) end up close.

Big rule

PEO number can NEVER be bigger than Employed number. Why? Because every PEO requires job first. No job = no PEO. So PEO ceiling = Employed %.

If Employed = 82.8%, no PEO card can show more than 82.8%.

Caveman: What is "As-of Year"?

Default = today's year (right now: **2026**).

This year decides which cohort each alumnus belongs to:

- Graduated in last 2 years → **Recent**
- Graduated 3–5 years ago → **Mid-Career**
- Graduated 6+ years ago (but not before 2018) → **Established**

It does NOT remove anyone. It just RELABELS them.

Caveman example

Alumnus graduated in 2023.

- Today (2026), $2026 - 2023 = 3$ years → **Mid-Career**.
- If you set as-of-year to 2025, then $2025 - 2023 = 2$ years → **Recent**.

Same alumnus. Different bucket. Because the question changed from "how does it look today?" to "how did it look last year?"

Why this control exists

1. **Old report rerun.** Accreditation people say "show us 2024 numbers." You set as-of-year to 2024. Boom — same data, but cohorts assigned the way they were back then.
2. **Frozen snapshot.** January 1 comes and suddenly people who used to be "Mid" become "Established." If you want a report that doesn't drift, lock as-of-year.
3. **Paper trail.** When you export PDF/Word, the cover page prints the as-of-year. So if reviewer asks later, "what year was this run as?" — answer is on page 1.

What it does NOT do

- Does NOT change PEO 1 / PEO 2 / PEO 3 percentages (those don't care about cohorts).
- Does NOT remove any alumni from the data (Year From / Year To do that).
- Cannot go before 2018 (database doesn't allow alumni earlier than that).

Caveman summary

Leave it blank = "right now." Set it = "pretend it's that year." Useful for old reports and frozen numbers.

Caveman: Cheat Sheet

PEO % can never beat Employed %.	(Job is required.)
Outcomes uses ALL alumni as bottom.	($447 \div 688$)
Field-Related uses ONLY employed as bottom.	($447 \div 570$)
Same top number. Different math.	
PEO 1/2/3 = "many doors to pass."	(Bigger top.)
Dashboard = "one door to pass."	(Smaller top, but smaller bottom too.)
As-of-year = relabel into cohorts.	Does not remove anyone.
Default = today's year.	
Set it = pretend it's that year.	